

1. OBJECTS

* Definition

- The object is something which have state, behaviour, identity and responsibility.
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• State-

- The attribute and value an object has
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• Behaviour

- Response given to outside world that an object can perform.
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• Identity

- Uniqueness associated with object

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• Responsibility

- Role of an object

- Role of an object

- Role of an object

- Role of an object

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- Role of an object

- Role of object
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* Examples of objects

1] Car

State - color, model, speed, fuel level
Behaviour - start, stop, accelerate, brake
Identity - car registration number
Responsibility - transport people safely

2] Fan

State - speed, number of blades, brand
Behaviour - rotate, change speed, turn on, off
Identity - fan serial number
Responsibility - circulate air

3] Mobile phone

State - battery, brand, OS, storage
Behaviour - call, message, charge, install app
Identity - IMEI number
Responsibility - communication and multimedia functions

4] Student

State - name, roll number, marks, class
Behaviour - study, write exam, submit assignment
Identity - roll number
Responsibility - learn and complete course

5. Book

State: title, author, pages, price

Behaviour: open, close, read

Identity: ISBN number

Responsibility: deliver information or story

6. Laptop:

State: RAM, Storage, OS, battery level

Behaviour: boot, shutdown, run program

Identity: laptop serial number

Responsibility: perform computing tasks

7. Refrigerator

State: temperature, capacity, brand

Behaviour: cool, freeze, auto defrost

Identity: serial number

Responsibility: preserve food

8. Teacher

State: Name, Subject, Experience

Behaviour: teach, check papers, grade

Identity: employee ID

Responsibility: deliver education

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~~ABSTRACTION~~ Encapsulation

Definition

- Binding and hiding of state & behaviour by default

- Bind - because they are inseparable in real life

* Example of Encapsulation

- Bank account hides balance and accesses only the deposit / withdrawal method
- Student marks stored privately and accessed through methods
- Login system hides password.
- Medical record stored securely accessed staff
- ATM pin protected inside banking system

4. INHERITANCE

- Reusing already defined class with a relationship.
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- Reusing already defined class with is a relationship.
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* Example of Inheritance.

- Dog - Animal
- Car - Vehicle
- Teacher - Person
- Manager - Employee
- Mobile ~~number~~ - Electronic Device
- Bus - Transport.
- Lion - Animal.
- Rectangle - Shape.
- Saving account - Bank account.

5. POLYMORPHISM

- Definition of Polymorphism
- Same message given to generalize thing same behaviour but implemented differently
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implemented differently *

Same message given to generalize thing same behaviour but implemented differently

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* Examples of polymorphism

- Shape draw

- calculator add

- Ride

- Message send

- payment

- Login authentication

- vehicle break

- vehicle speed

- [illegible]

Book

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* Examples of ~~Abstraction~~ Abstraction

- 1] ATM machine.
- 2] car driving
- 3] Mobile phone
- 4] TV- remote
- 5] Washing machine
- 6] Google Search
- 7] Elevator
- 8] online payment
- 9] Electric Switch
- 10] Computer OS.